Does Syntactic Structure Make Language Models More Robust to Noise?

LSTM vs. RNNG Subject–Verb Agreement Under Training-Data Corruption

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Outline

Motivation: Can grammar survive noisy input?

- Children acquire robust grammatical systems despite hearing noisy and imperfect language
- This observation motivates the classic **Poverty of the Stimulus** debate:
  Is grammar learned from data, or supported by innate structure?
- Human experiments cannot systematically manipulate children's linguistic input
- Neural language models provide a controllable testbed:
  - architecture = innate structure
  - training corpus = linguistic environment

Research Questions

RQ1: Can grammar be learned from noisy input?

- How much grammatical corruption can a model tolerate?
- Is there a critical noise threshold beyond which acquisition breaks down?

RQ2: How does noise change generalization?

- Does the model maintain a hierarchical syntactic rule?
- Or does it fall back on a simpler linear nearest-noun heuristic?

RQ3: Does structure buy robustness?

- Compare LSTM and RNNG under identical noise conditions.
- Does explicit syntactic bias help preserve grammatical learning?

Subject–verb agreement: hierarchical vs. linear

- **Hierarchical rule**: agree with the true syntactic subject.
- **Linear rule**: agree with the noun closest to the verb.

Example

The keys to the cabinet _____ *on the table.*

- True subject = *keys* (plural)
- Nearest noun = *cabinet* (singular)

- Hierarchical ⇒ **are**
- Linear ⇒ **is**

Key idea: SVA with attractors creates cases where the two rules make *different* predictions — our diagnostic lever.

Attractors as the diagnostic case

- An **attractor** is an intervening noun that mismatches the subject in number (e.g. *cabinet* between *keys* and the verb).
- On **no-attractor** items, both rules agree $\Rightarrow$ accuracy is high for either strategy.
- On **attractor** items, only the hierarchical rule is correct $\Rightarrow$ success requires ignoring the misleading linear cue.
- The **attractor gap** (no-attractor acc $-$ attractor acc) measures how much the model relies on the linear shortcut.

Architectural inductive biases

Two competing learners

LSTM

- Sequential recurrent model
- No explicit syntactic structure
- Learns grammar from statistical regularities

Weak structural prior

RNNG

- Generates sentences and parse trees jointly
- Explicit hierarchical representations
- Strong syntactic inductive bias

Strong structural prior

Key question: Does explicit syntactic structure help preserve hierarchical rules under noisy input?

Related work and research gap

SVA as a syntax benchmark

- Linzen et al. (2016): LSTMs learn agreement but struggle with attractors.
- Gulordava et al. (2018): evidence for abstract syntactic generalization.

Structural inductive bias

- Dyer et al. (2016): Recurrent Neural Network Grammar (RNNG).
- Kuncoro et al. (2018): explicit syntax improves long-distance dependencies.

Research gap

- Prior work tests models on clean data; little asks whether grammatical rules can still be *acquired* under noisy supervision.
- No systematic study of how SVA corruption affects rule acquisition, hierarchical vs. linear generalization, or LSTM vs. RNNG robustness.

Noise injection: five corrupted corpora

- Base: Linzen et al. (2016) SVA corpus — $\sim$1.58M Wikipedia sentences, vocab capped at the 10K most frequent types.
- Each sentence annotated with subject, verb, verb position, and # intervening attractor nouns.
- Build five training corpora, identical except agreement noise: flip the verb's number with probability p.

$$p \in \{0,\ 0.004,\ 0.02,\ 0.10,\ 0.50\} \quad \text{(baseline} \rightarrow \text{high)}$$

- Flip via deterministic inflection (`lemminflect` + irregular copulas/auxiliaries); all other lexical/syntactic content preserved.
- Training-set size held constant $\Rightarrow$ isolates noise.

Leak-free evaluation protocol

- The five corpora are row-aligned with the source corpus.
- Reserve one fixed held-out set: 20k sentence indices (seed 1234).
- Those rows excluded from **every** training corpus $\Rightarrow$ no test leakage under any noise condition.
- Test items built from the **uncorrupted** source $\Rightarrow$ all test sentences are grammatical.
- Same test set for all conditions — only training noise varies.

Minimal-pair test

- For each held-out sentence: a prefix, the grammatical verb v^+, and its number-flipped counterpart v^-.
- Model is **correct** iff it prefers the grammatical form:

$$P(v^+ \mid \text{prefix}) \; > \; P(v^- \mid \text{prefix})$$

- Each pair tagged `has_attractor`: an intervening noun mismatching the subject in number.

Models

- **Sequential baseline** — word-level LSTM LM (Linzen 2016 / Gulordava 2018 style): 2 layers, tied embeddings, dropout; trained from scratch per condition.
- **Structurally-biased** — RNNG (Dyer et al. 2016): explicit hierarchical structure.
- Core question: does a structural bias make hierarchical agreement *more robust* to noisy supervision?
- Fallback: ON-LSTM (Shen et al. 2019) — softer structural bias, no parse supervision needed.

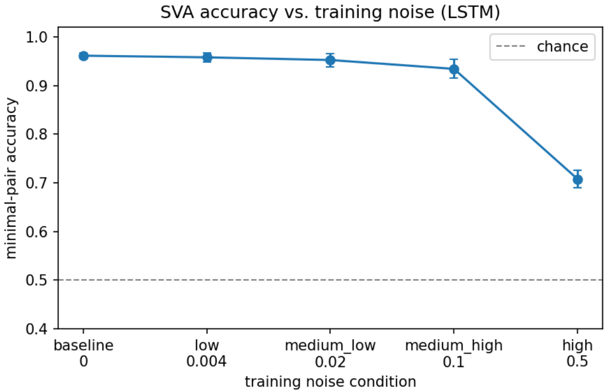
Metrics and rule probing

- Primary: minimal-pair **accuracy** (grammatical $>$ ungrammatical).
- Secondary (graded): **surprisal** of the correct verb, $s = -\log_2 P(v^+ \mid \text{prefix})$.
- Split accuracy by attractor presence; reliance on the linear cue = **attractor gap**

$$\Delta = \text{acc}_{\text{no-attr}} - \text{acc}_{\text{attr}}$$

- Small $\Delta \rightarrow$ hierarchical rule; large $\Delta \rightarrow$ nearest-noun (linear) rule.
- Averaged over seeds $\{1, 2, 3\}$; central analysis: Δ vs. noise.

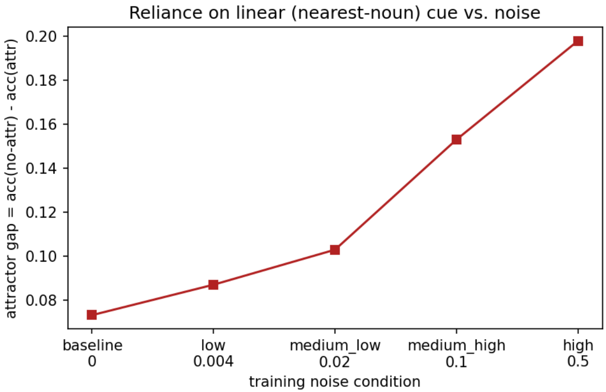
Accuracy degrades with noise — with a threshold



- Flat up to 10% noise: 0.961 → 0.934
- Collapses at 50%: down to 0.708
- Tolerates ∼1 bad verb in 10, then breaks

⇒ critical-threshold effect, not gradual decay

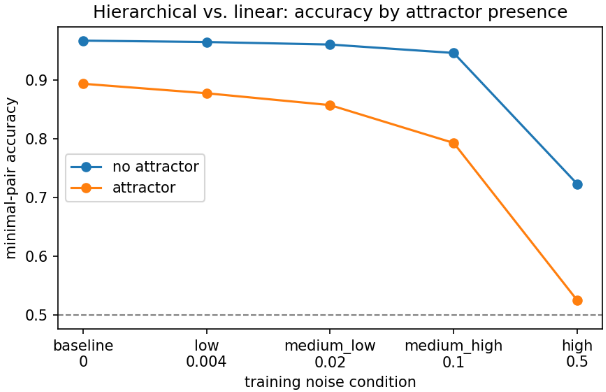
The attractor gap widens with noise



- Gap $=$ acc(no-attractor) $-$ acc(attractor)
- Grows monotonically: $0.073 \to 0.198$
- Index of reliance on the *linear* nearest-noun rule

$\Rightarrow$ noise pushes the model toward the shortcut

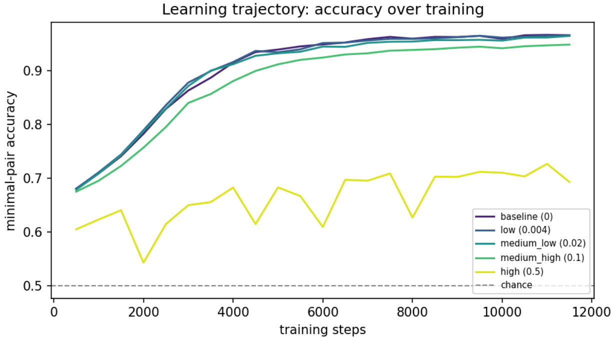
Accuracy by attractor presence



- No-attractor items stay > 0.94 until 50%
- Attractor items fall to near chance (0.525)
- The two rules disagree only on attractor items

$\Rightarrow$ that is exactly where noise bites first

Learning trajectories: delay vs. prevention



- $\leq 10\%$ noise: same endpoint, just slower $\rightarrow$ **delay**
- 50% noise: plateaus at ~ 0.69, never learns $\rightarrow$ **prevention**
- Clean: gap shrinks with training $(0.15 \rightarrow 0.05)$
- 50%: gap *grows* with training $(0.14 \rightarrow 0.20)$

When does the rule break?

- **Threshold, not gradual**: absorbs noise up to 10%, snaps at 50% — like a child's grammar surviving occasional speech errors.
- **Attractor gap is the early warning**: it climbs ($0.07 \rightarrow 0.20$) while overall accuracy still looks healthy.
- **Delay vs. prevention**: low noise only *delays* acquisition (curves converge); 50% noise *prevents* it and entrenches the linear rule.

Does structure buy robustness?

- The headline question — does a syntactic prior (RNNG) resist noise better than a sequential LSTM? — needs the RNNG runs (in progress).
- **LSTM baseline**: no explicit hierarchy bias $\Rightarrow$ learns the rule cleanly when input is clean, but drifts to the linear shortcut as noise grows.
- If RNNG holds the hierarchical rule where the LSTM has caved $\Rightarrow$ structure buys robustness.
- If both fail together $\Rightarrow$ the bottleneck is input quality, not architecture — speaking to poverty-of-the-stimulus.

Takeaways and future work

Takeaways

- LSTMs tolerate moderate SVA noise but have a sharp breaking point.
- Noise trades the hierarchical rule for the linear nearest-noun heuristic.
- Heavy noise prevents acquisition outright — more training makes it worse.

Limitations & next steps

- Synthetic, uniform noise; templatic corpus; single-seed trajectories.
- Complete the RNNG arm; try ON-LSTM as a no-gold-tree syntactic baseline.
- Move toward naturalistic, frequency-dependent noise and larger scale.

Thank you!

Questions?